PHYS — Semiconductor Physics

Long-Form Exercise 2 (Ch. 5–6 Focus)

Submission:

- Submit a single PDF report (exported from Markdown/Word/LaTeX).
- Submit your **code as a .zip file** (Python/Matlab/etc.) separately through Canvas.
- All plots must be generated computationally. No hand-drawn plots.
- Axes must be annotated with units.

Structure:

- Each section begins on a new page.
- After each computational result, include a **Reflection/Discrepancy Analysis** subsection.
- Final day: **Presentation** (10–15 min per student).

Task 1 — Drift and Diffusion Currents

- Implement the drift and diffusion current densities:

$$J_{\text{drift}} = qn\mu E, \qquad J_{\text{diff}} = qD\frac{dn}{dx}.$$

- Plot J_{drift} vs. E for electron and hole mobilities.
- Plot J_{diff} for a linear carrier gradient.
- Verify the **Einstein relation**:

$$\frac{D}{\mu} = \frac{k_B T}{q}.$$

Task 2 — Mobility and Temperature Dependence

- Implement a model $\mu(T) \propto T^{-m}$, where m reflects dominant scattering.
- Compute and plot $\mu(T)$ over $200\,\text{K} \leq T \leq 600\,\text{K}$.
- Compare electrons vs. holes in your assigned semiconductor.
- Reflect on how scattering mechanisms (phonons, impurities) influence the result.

Task 3 — Nonequilibrium Excess Carriers

- Solve the continuity equation for excess carriers under steady-state illumination:

$$\frac{\partial \Delta n}{\partial t} = G - \frac{\Delta n}{\tau}.$$

- Plot $\Delta n(t)$ showing recombination with carrier lifetime τ.
- Explore at least two cases: **short lifetime** vs. **long lifetime**.
- Compare your results with literature values of lifetimes.

Task 4 — Quasi-Fermi Levels

- For a forward-biased junction, compute quasi-Fermi levels E_{Fn} and E_{Fp}.
- Plot E_{Fn} and E_{Fp} relative to E_c and E_v.
- Discuss the physical meaning:
  - Separation $\Delta E_F = qV_a$ corresponds to applied bias.

Task 5 — Experimental Comparison

- Find at least one dataset (mobility, diffusion constant, or lifetime) for your semiconductor.
- Overlay with your computational results.
- Identify one **agreement** and one **discrepancy**.
- Suggest at least one reason for mismatch.

Reflection and Discrepancy Analysis (15%)

Each section must include a short reflection.

Template

1. **What did you expect?**
   (Describe the anticipated trend or limit.)
2. **What did you find?**
   (Summarize the computed result.)

3. **If there's a mismatch, why?**
   (Give at least one explanation in 2–3 sentences.)

Grading Breakdown

- Computational tasks and plots: **45%**
- Literature comparison: **15%**
- Reflection and discrepancy: **15%**
- Presentation: **25%**

Report Format

Your report must be a single PDF with the following structure:

1. Introduction

- Identify your assigned semiconductor.
- State why transport and recombination are important.
- Outline the scope: drift, diffusion, mobility, lifetimes, quasi-Fermi levels.

2. Methods

- Present equations used (with sources).
- Note assumptions (e.g., constant effective mass, uniform doping).
- Include short code snippets where relevant.

3. Results

- Drift/diffusion current plots
- $\mu(T)$ vs. T
- $\Delta n(t)$ for two lifetimes
- Quasi-Fermi level separation plot
- Experimental comparison overlay

4. Reflection and Discrepancy Analysis

- Follow the template for each result.

5. Discussion

- Summarize insights: role of mobility, lifetimes, quasi-Fermi levels.
- Compare results to literature.
- Note limitations of your approach.

6. Conclusion

- Restate your key findings.
- Identify one thing you learned from coding.
- State one open question about your material.

7. References

- IEEE style.
- Include at least **two sources** (textbook + journal).
- Submit PDFs of cited references in your archive.

Midterm Presentation Format (10–15 min)

1. **Introduction** — Semiconductor and why it matters.
2. **Methods** — Equations, assumptions, code structure.
3. **Key Results** — Drift/diffusion, mobility, lifetime plots, quasi-Fermi levels.
4. **Experimental Comparison** — Overlay + discussion.
5. **Reflection** — One agreement, one discrepancy, one limitation.
6. **Conclusion** — One next step.
7. **Live Demo** — Re-run one plot (e.g., $\mu(T)$).

Academic Integrity

- References must be real and included as PDFs.
- AI-generated or fabricated citations will be penalized.
- You may use coding help (AI, StackOverflow, etc.) but must **explain your methods** in your own words.